

# Technical Document #873: Retrieval Augmented Generation - Comprehensive Overview

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Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

### Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Hybrid search combines keyword-based and semantic search.